

* The original value of a statistical life was calculated in 1990\$. We have used a factor of 1.8134, based on the All-Items CPI-U.

H.2 Hospital Admissions & Emergency Room Visits

This section presents the core values for avoided hospital admissions, as well as avoided emergency room visits. We assume that hospital admissions due to acute exposure to air pollution pass through the emergency room. However, the value of hospital admissions that we have calculated here does not account for the cost incurred in the emergency room visit.

H.2.1 Hospital Admissions

As suggested above, the total value to society of an individual's avoidance of a hospital admission can be thought of as having two components: (1) the cost of illness (COI) to society, including the total medical costs plus the value of the lost productivity, as well as (2) the WTP of the individual, as well as that of others, to avoid the pain and suffering resulting from the illness.

In the absence of estimates of social WTP to avoid hospital admissions for specific illnesses (components 1 plus 2 above), estimates of total COI (component 1) are available for use in BenMAP as conservative (lower bound) estimates. Because these estimates do not include the value of avoiding the pain and suffering resulting from the illness (component 2), they are biased downward. Some analyses adjust COI estimates upward by multiplying by an estimate of the ratio of WTP to COI, to better approximate total WTP. Other analyses have avoided making this adjustment because of the possibility of over-adjusting -- that is, possibly replacing a known downward bias with an upward bias. Based on Science Advisory Board (SAB) advice, the COI values currently available for use in BenMAP are not adjusted.

Unit values are based on ICD-code-specific estimated hospital charges and opportunity cost of time spent in the hospital (based on the average length of a hospital stay for the illness). The opportunity cost of a day spent in the hospital is estimated as the value of the lost daily wage, regardless of whether or not the individual is in the workforce.

For all hospital admissions endpoints available in BenMAP, estimates of hospital charges and lengths of hospital stays were based on discharge statistics provided by the Agency for Healthcare Research and Quality's Healthcare Utilization Project National Inpatient Sample (NIS) database (2016). The NIS is the largest inpatient care database in the United States, and it is the only national hospital database containing charge information on all patients. It contains data from a very large nationally representative sample of about eight million hospital discharges, and therefore provides the best estimates of mean hospital charges and mean lengths of stay available, with negligible standard errors. The sampling frame for the 2016 NIS is a sample of hospitals that comprises approximately 90 percent of all hospital discharges in the United States. Since the NIS is based on discharge samples, the discharge-level weight was used to weight discharges in order to produce national estimates. The principal diagnoses were used to define the health endpoints.

Since most pollution-related hospital admissions are likely unscheduled, the unit values of avoided hospital admissions used in BenMAP are based solely on unscheduled hospitalizations. The total COI for an ICD-code-specific hospital stay lasting n days is estimated as the mean hospital charge plus n times the daily lost wage.

County-specific median annual income divided by (52×5) was used to estimate county-specific median daily wage. The data source for median annual income is the 2021 American Community Survey (ACS). ACS provided data for median annual income for all individuals over 16 years old in all U.S. counties. We apply county-specific employment rates from the 2021 ACS to the mean wage data to estimate the average economic cost of a lost day or year for non-workers and for workers. For non-workers, the daily lost wage is the ACS mean daily wage net of a seven percent income tax rate, which is the percentage difference in median post-tax income and median income from U.S. Census Bureau (2021). For workers, the daily lost wage is the ACS mean daily wage scaled by a multiplier that accounts for mean county-level fringe benefits and overhead. We calculate this value using county-specific fringe benefits estimates from the U.S. Bureau of Labor Statistics' 2023 National Compensation Survey and a national overhead coefficient of 1.2 from U.S. EPA's National Center for Environmental Economics (NCEE), as described in U.S. EPA (2023). Because wage data used in BenMAP are county-specific, the unit value for a hospital admission varies from one county to another.

Although the data for hospital charges are from year 2016, the default hospital admission unit values in BenMAP are in year 2015 dollars to be consistent with the unit values of other health endpoints in BenMAP. This was done by inflating the medical costs (2016 dollars) to 2015 dollars using BenMAP's medical inflation index.

The hospital admission outcomes that the EPA uses in its regulatory analyses are given in Table H-3. Although unit values available for use in BenMAP are county-specific, the national median daily wage was used to calculate opportunity costs and total costs.

Table H-3. Core Unit Values Available for Hospital Admissions

Endpoint	ICD Codes	Age Range		Mean Hospital Charge (2015 \$)	Mean Length of Stay (days)	Total Cost of Illness (Unit Value in 2015\$)*
		Min	Max			
HA, All Cardiac Outcomes	390-459	0	99	\$16,045	5.05	\$16,918
HA, All Respiratory	460-519	0	18	\$9,075	3.49	\$9,678
HA, All Respiratory	460-519	65	99	\$35,402	6.07	\$36,451
HA, Alzheimer's Disease	331.0	65	99	\$10,696	7.95	\$12,070
HA, Cardio-, Cerebro- and Peripheral Vascular Disease	410- 414, 429, 426- 427, 428, 430-438, 440-449	65	99	\$14,665	4.82	\$15,498
HA, Respiratory-1	491, 492, 493, 496	0	99	\$7,676	3.86	\$8,343
HA, Respiratory-2	464-466, 480-487, 490-492, 493	65	99	\$9,003	4.66	\$9,808
HA, Parkinson's Disease	332	18	99	\$12,190	3.83	\$12,852

* The opportunity cost of a day spent in the hospital was estimated, for the above exhibit, at the median daily wage of all workers, regardless of age. The median daily wage was calculated by dividing the median weekly wage (\$864 in 2015\$) by 5. The median weekly wages for 2015 were obtained from the U.S. Census Bureau's 2015 American Community Survey, "Selected Economic Characteristics: 2015 American Community Survey 1-Year Estimates."

For two hospital admission endpoints, Alzheimer's and Parkinson's Disease, we calculated the lifetime cost of illness in addition to calculating the costs associated with the initial hospitalization.

Valuation sources of Alzheimer's disease lifetime medical costs were available from the Alzheimer's Association (2020) report and Jutkowitz et al. (2017). Using Alzheimer's Association (2020), we first developed an estimate of incremental annual medical expenses for Medicare beneficiaries living with Alzheimer's Disease. Then, using the estimated life expectancy duration of 5 years from Jutkowitz et al. (2017), we estimated total discounted present values for a five-year stream of costs using 2, 3 and 7 percent discount rates (Table H-4). We note that the median age of Alzheimer's disease onset is after the age of 65. As such, we exclude any potential lost wages given the low rate of labor force participation in this age group. Lifetime medical costs, excluding initial hospitalization, are estimated at \$159,934 using a 2% discount rate, \$156,920 using a 3% discount rate or \$145,946 using a 7% discount rate in 2015.

Table H-4. Alzheimer's Disease Valuation (2015\$)

Year	2% Discount Rate	3% Discount Rate	7% Discount Rate
0	\$33,266	\$33,266	\$33,266
1	\$32,614	\$32,297	\$31,090
2	\$31,974	\$31,357	\$29,056
3	\$31,347	\$30,443	\$27,155
4	\$30,733	\$29,557	\$25,379
Total Lifetime Costs	\$159,934	\$156,920	\$145,946

Estimates of lifetime costs for Parkinson's Disease were provided by Yang et al. (2020), including direct, indirect, and non-medical costs. Using Yang et al. (2020), we first developed an annual estimate of excess costs associated with living with Parkinson's Disease. Then, using the estimated life expectancy duration of 14.6 years from De Pablo-Fernandez et al., 2017, we calculated the present value of lifetime costs over this period using 2, 3 and 7 percent discount rates (Table H-5). Lifetime medical costs are estimated at \$571,886 using a 2% discount rate, \$537,409 using a 3% discount rate, and \$428,398 using a 7% discount rate in 2015\$.

Table H-5. Lifetime Parkinson's Disease Valuation Estimate Calculation

Year	2% Discount Rate	3% Discount Rate	7% Discount Rate
0	\$44,718	\$44,718	\$44,718
1	\$43,841	\$43,416	\$41,793

Year	2% Discount Rate	3% Discount Rate	7% Discount Rate
2	\$42,982	\$42,151	\$39,059
3	\$42,139	\$40,924	\$36,503
4	\$41,313	\$39,732	\$34,115
5	\$40,502	\$38,574	\$31,883
6	\$39,708	\$37,451	\$29,798
7	\$38,930	\$36,360	\$27,848
8	\$38,166	\$35,301	\$26,026
9	\$37,418	\$34,273	\$24,324
10	\$36,684	\$33,275	\$22,732
11	\$35,965	\$32,305	\$21,245
12	\$35,260	\$31,364	\$19,855
13	\$34,568	\$30,451	\$18,556
13.6	\$19,671	\$17,117	\$9,942
Total Lifetime Costs (14.6 yr survival)	\$571,886	\$537,409	\$428,398

H.2.2 Emergency Room Visits

As with hospital admissions, to value emergency room visits we develop primary COI estimates using data from the Healthcare Cost and Utilization Project (HCUP). The 2016 Nationwide Emergency Department Sample (NEDS) provides recent, nationally representative information on medical treatment in emergency departments. In the case of emergency department visits, valuation estimates include only the medical costs.

The NEDS dataset includes discharge-level observations. That is, each data point represents one individual being discharged from the emergency department (NEDS). Because individuals are treated in these settings for a variety of reasons, we use medical billing codes to extract observations related to each health endpoint. The epidemiological studies described in Appendix E, F and G provide ICD-9 codes for each illness; however, recent HCUP datasets (including NEDS) use ICD-10 codes. Thus, we first crosswalk the relevant ICD-9 codes to associated ICD-10 codes using a mapping provided by the U.S. Centers for Disease Control. We then identify all discharges in the HCUP datasets with ICD-10 codes that match to a study's ICD-9 code(s). Because HCUP datasets often include multiple ICD-10 codes for each discharge, we focus on the principal diagnosis (i.e., the first-listed ICD-10 code).

In the NIS dataset, we convert total charges (i.e., the amount billed to patients, employers, or insurance providers) into estimates of total costs (i.e., the final reimbursements for medical treatment). Unadjusted charges are not suitable for use in regulatory analysis because posted prices generally do not reflect actual medical costs due, in part, to negotiation between medical providers and payers (e.g., insurance companies). We assume that adjusted charges reflect the actual revenue the hospital